Capture the Flag 7: Stop Putting Everything on One Screen

Goal: Split your Student Directory into multiple screens using Flutter navigation, while using a modal for editing student information.

Instructor Note
These activities require your own problem-solving and research skills.
DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

🏴 Flag 0 — Create Your Feature Branch

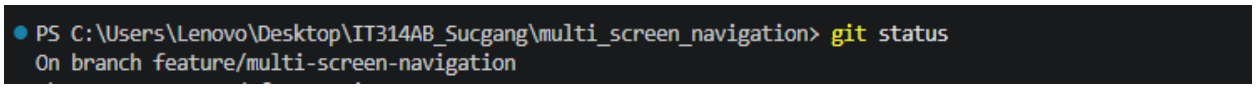
Objective: Create a new feature branch before adding multiple screens.

Create:

`feature/multi-screen-navigation`

Screenshot:

- Terminal showing your current branch



🏴 Flag 1 — Create the Student List Screen

Objective: Move your Student Directory into its own screen.

Create a dedicated **Student List Screen** containing your existing student list.

Research how Flutter creates separate screens using `Widget` classes.

Test

The Student Directory should function as its own screen.

Screenshot:

- Student List Screen
- Student List screen code
- Application showing the screen

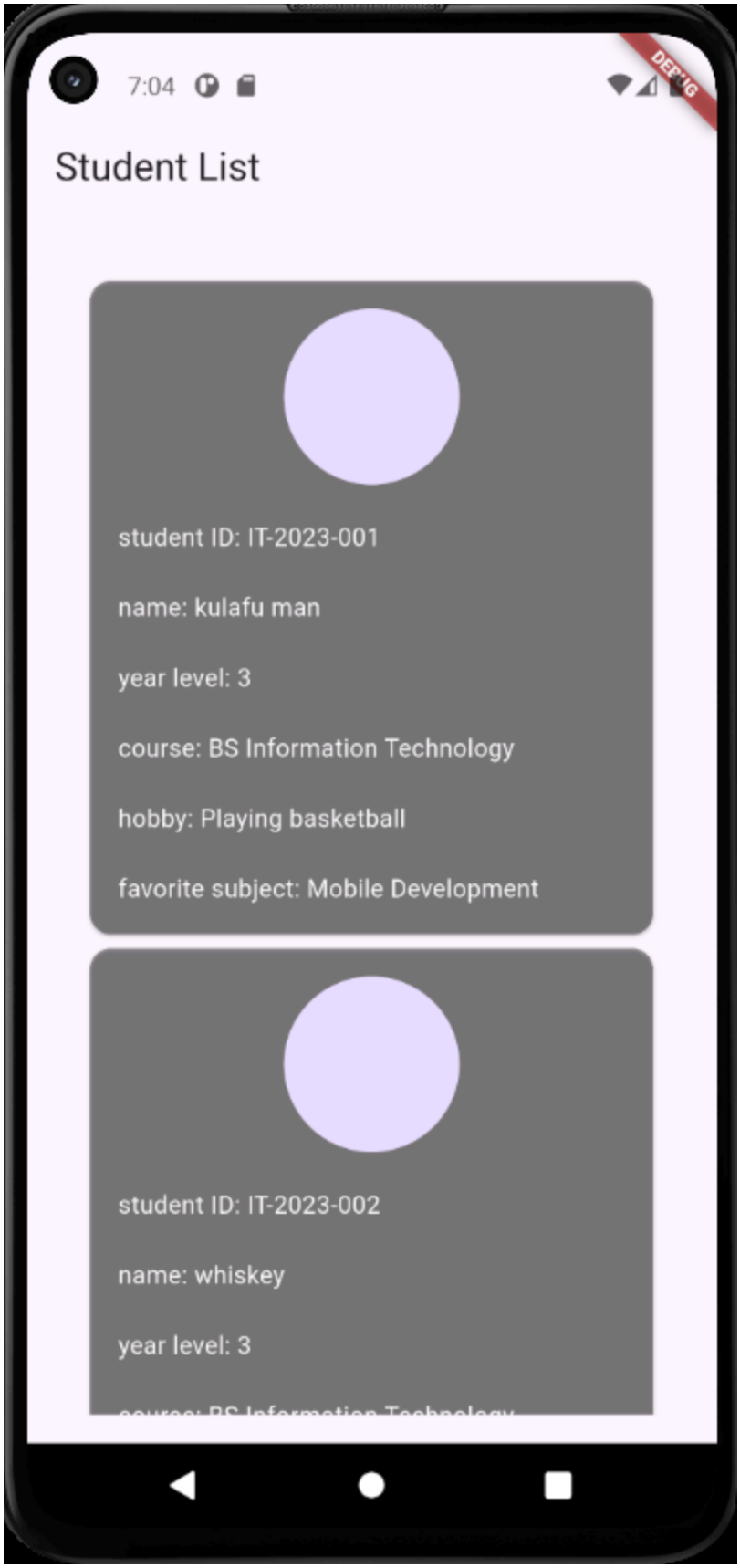
```
  appBar: AppBar(title: const Text('Student List')),
  body: Row(children: [StudentListScreenW()]),
), // Scaffold
```

```
    favoriteSubject: 'Object-Oriented Programming',
  ),
  Student(
    studentId: 'IT-2023-010',
    name: 'hatdog hans',
    yearLevel: 3,
    course: 'BS Information Technology',
    hobby: 'Creative writing',
    favoriteSubject: 'Information Assurance and Security',
  ),
];

class StudentListScreenW extends StatefulWidget {
  const StudentListScreenW({super.key});

  @override
  State<StudentListScreenW> createState() => _StudentListScreenWState();
}

class _StudentListScreenWState extends State<StudentListScreenW> {
  @override
  Widget build(BuildContext context) {
    return Container(child: Text("StudentList widget"));    Unnecessary instance of 'Container'.↵Try removing the 'Container' (b
  }
}
```


7:04
DEBUG
Student List
student ID: IT-2023-001
name: kulafu man
year level: 3
course: BS Information Technology
hobby: Playing basketball
favorite subject: Mobile Development
student ID: IT-2023-002
name: whiskey
year level: 3

🏴 Flag 2 — Create the Student Details Screen

Objective: Create a separate screen for viewing a student's information.

When a student is selected, navigate:

Student List → Student Details

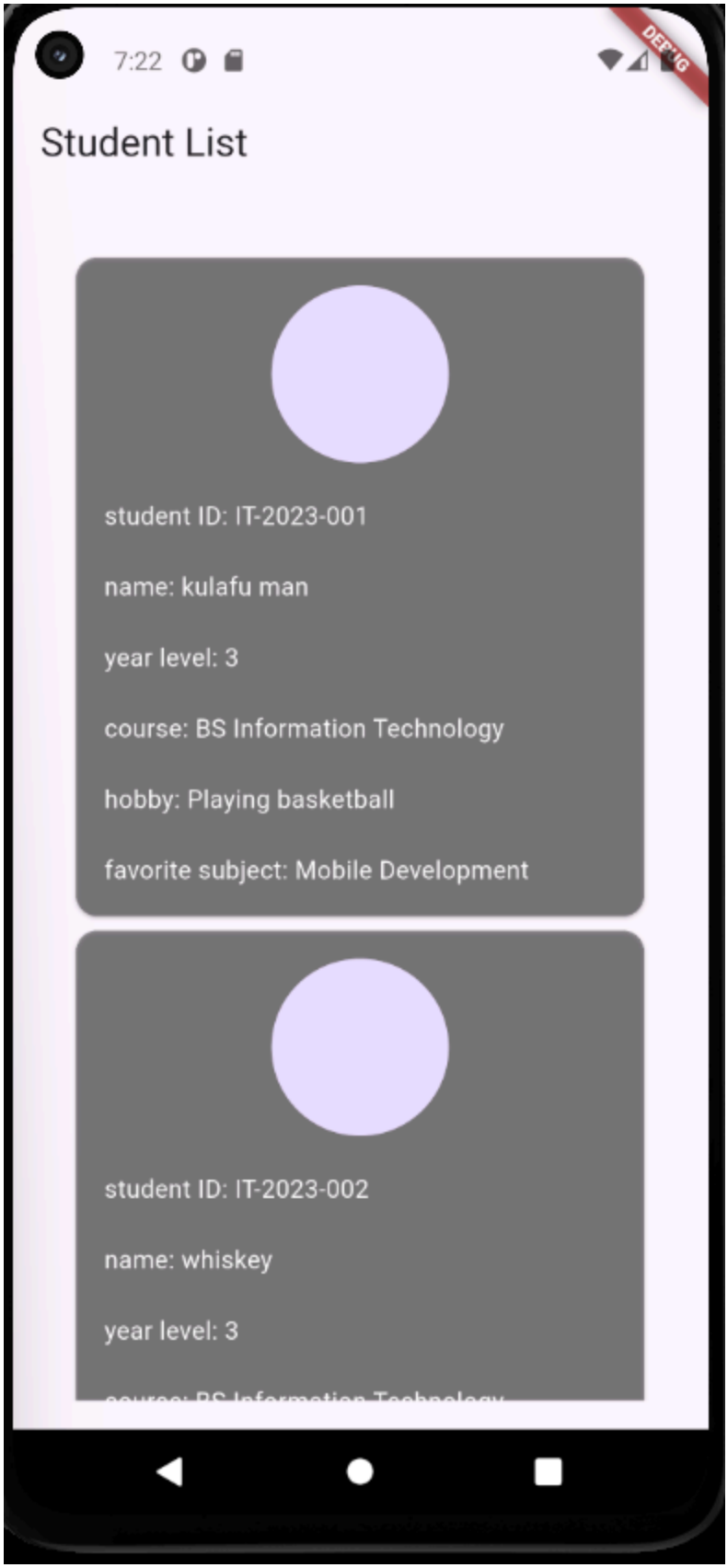
Research Flutter's `Navigator` and how to move between screens.

Test

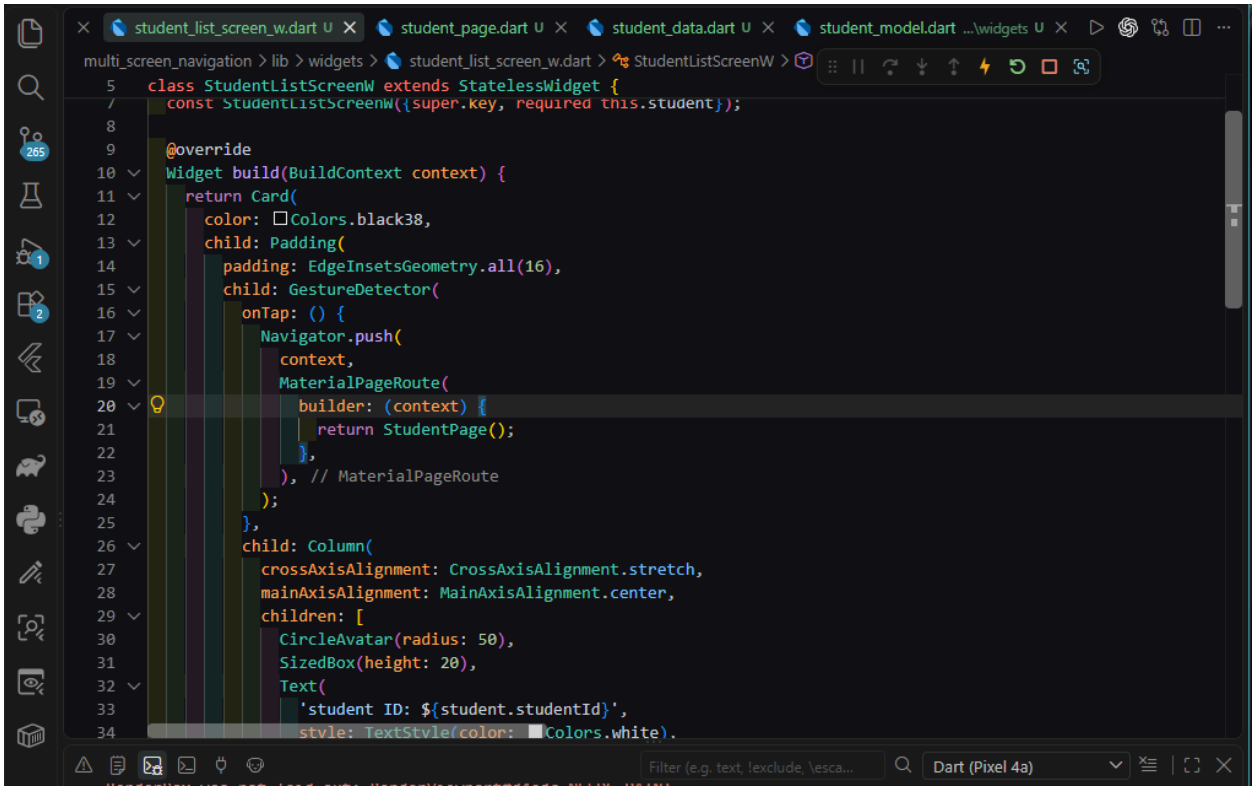
1. Select a student.
2. Open the Student Details Screen. (No data displayed yet)
3. Return to the Student List using back navigation.

Screenshot:

- Student List Screen
- Student Details Screen
- Navigation code


7:22
DEBUG
Student List
student ID: IT-2023-001
name: kulafu man
year level: 3
course: BS Information Technology
hobby: Playing basketball
favorite subject: Mobile Development
student ID: IT-2023-002
name: whiskey
year level: 3


7:22
DEBUG
student information
under construction



🏴 Flag 3 — Pass Student Data

Objective: Pass the selected student's data to the Student Details Screen.

The Details Screen should display the correct information for the selected student.

Display relevant information such as:

- Name
- Student ID
- Course
- Status
- Other existing student data

Research how data can be passed between Flutter screens.

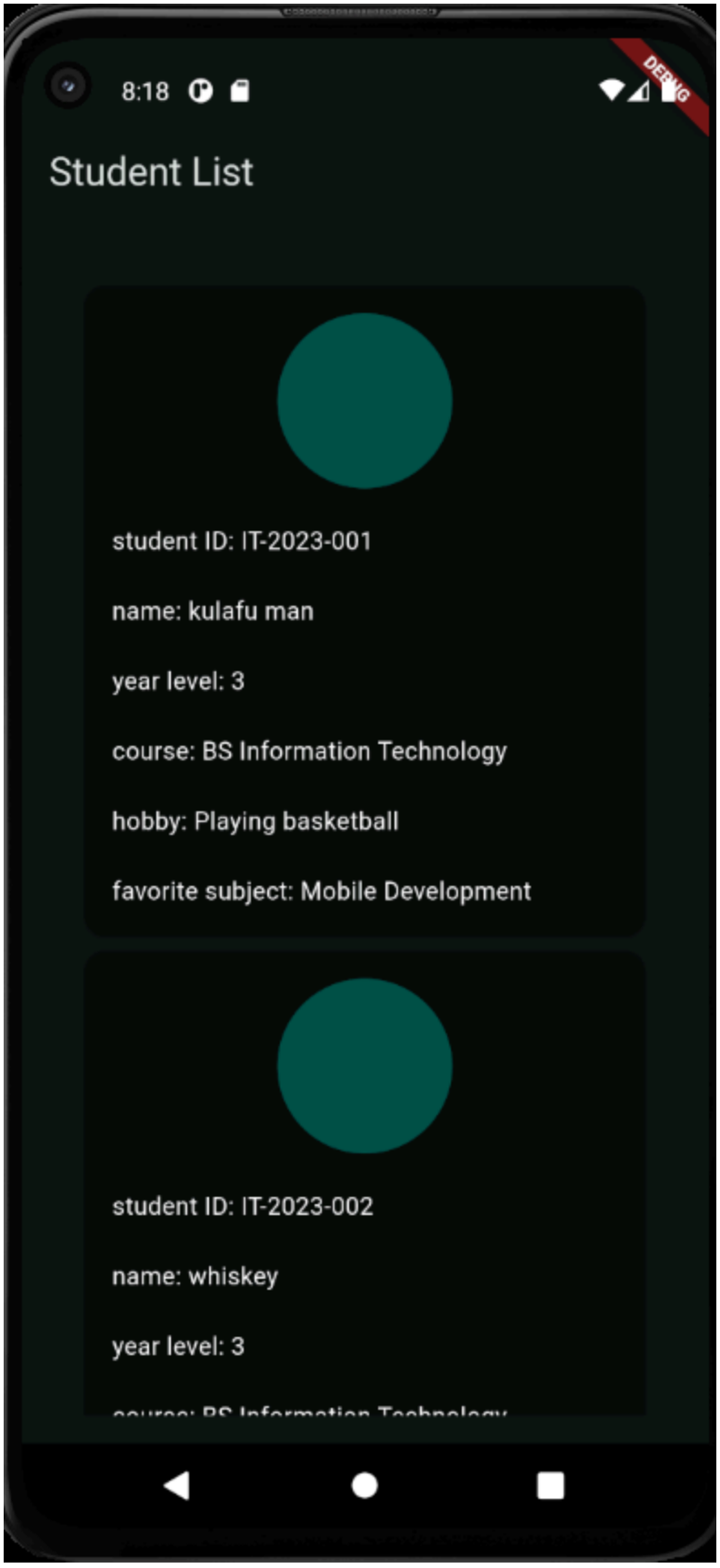
Test

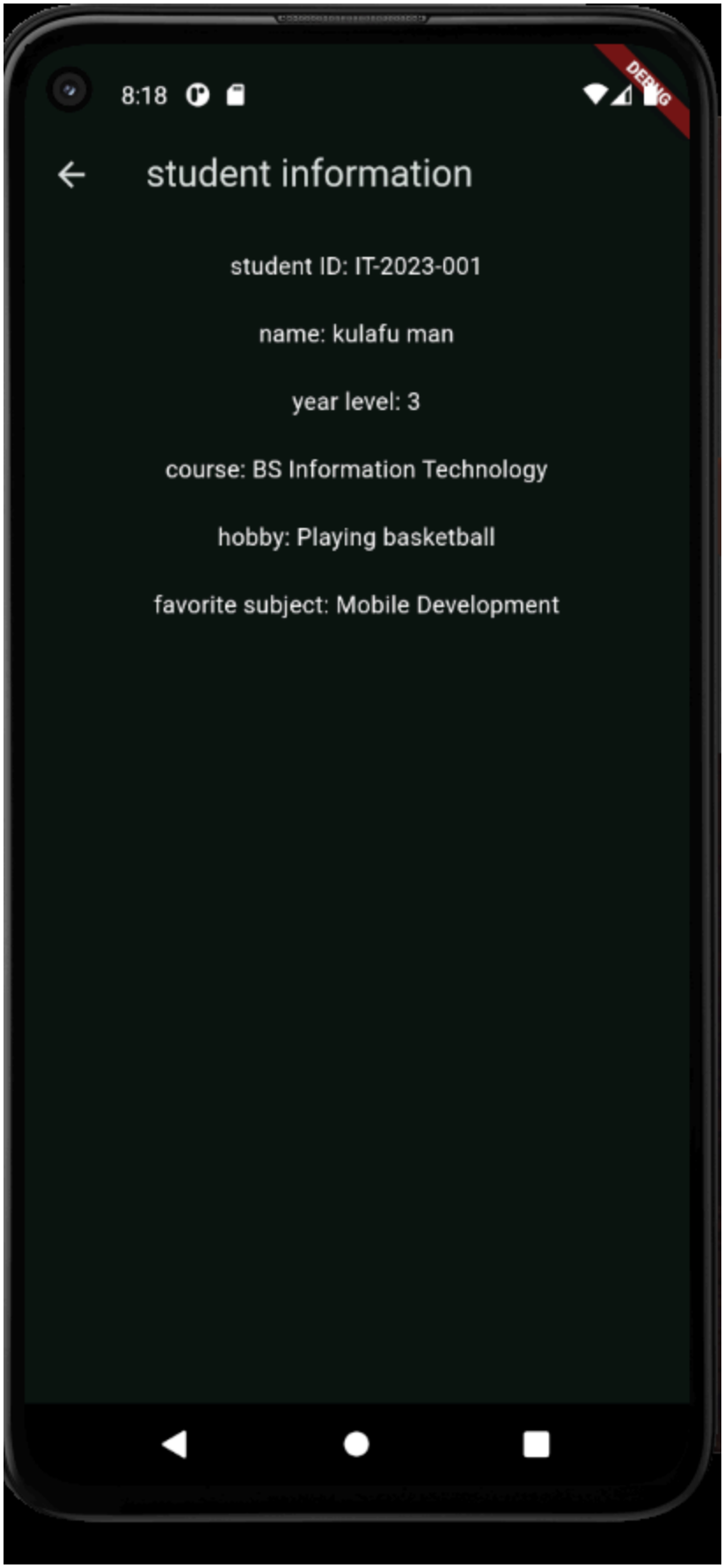
Selecting different students should display their corresponding information.

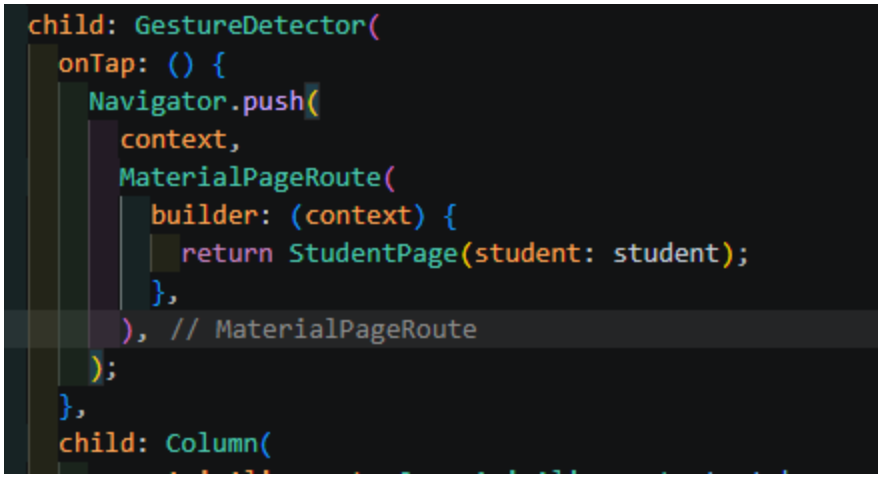
Student A → Student A's Details
Student B → Student B's Details

Screenshot:

- Selected student
- Student Details
- Data-passing code


8:18
DEBUG
Student List
student ID: IT-2023-001
name: kulafu man
year level: 3
course: BS Information Technology
hobby: Playing basketball
favorite subject: Mobile Development
student ID: IT-2023-002
name: whiskey
year level: 3


8:18
DEBUG
student information
student ID: IT-2023-001
name: kulafu man
year level: 3
course: BS Information Technology
hobby: Playing basketball
favorite subject: Mobile Development



```
child: GestureDetector(
  onTap: () {
    Navigator.push(
      context,
      MaterialPageRoute(
        builder: (context) {
          return StudentPage(student: student);
        },
      ), // MaterialPageRoute
    );
  },
  child: Column(
```

🏴 Flag 4 — Edit with a Modal

Objective: Allow administrators to edit student information without creating another screen.

Add an **Edit** action to the Student Details Screen.

When selected, display an editing interface using a **modal**, such as a dialog or bottom sheet.

Research Flutter's:

- `showDialog`
- `showModalBottomSheet`

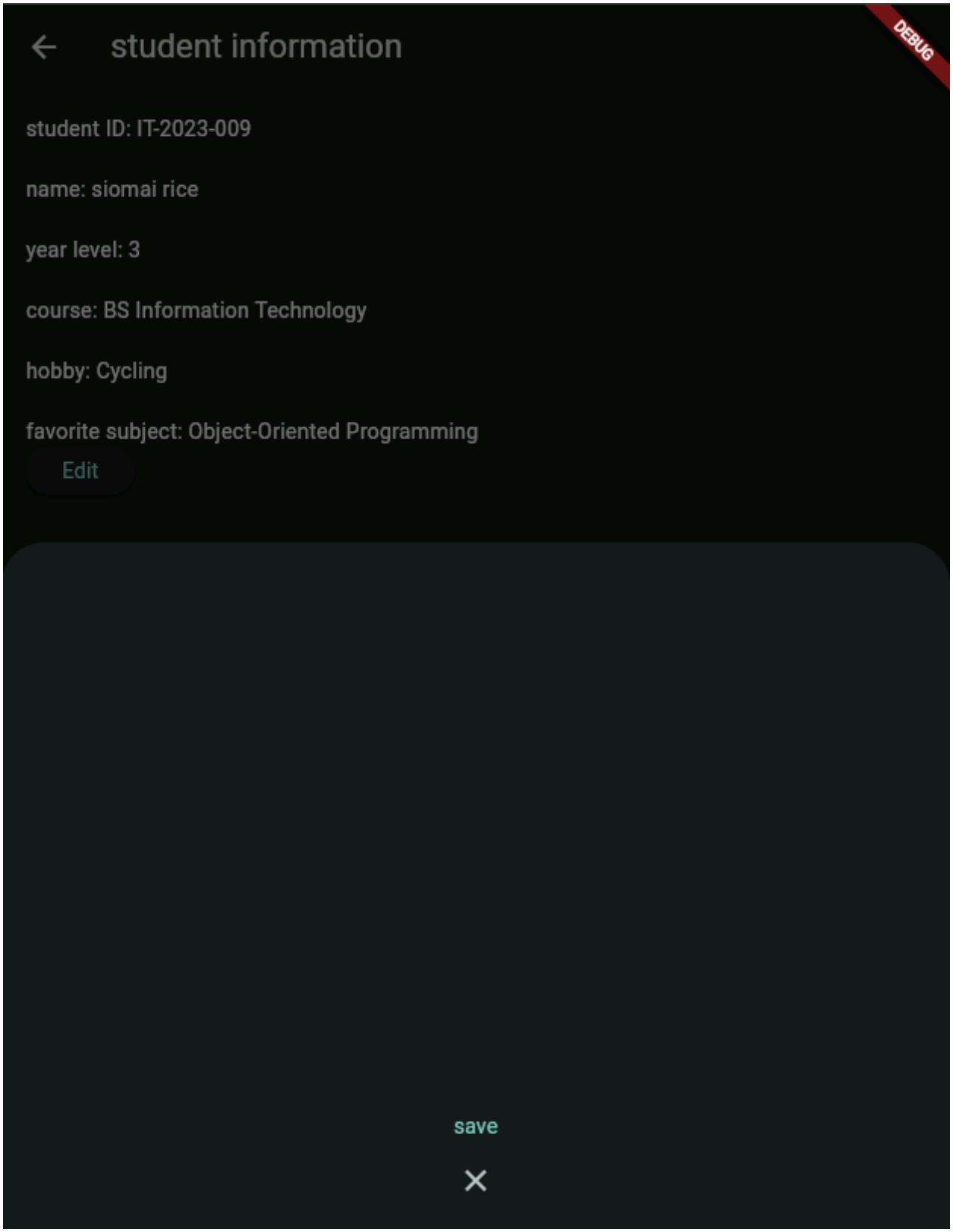
You may choose whichever approach is appropriate for your design.

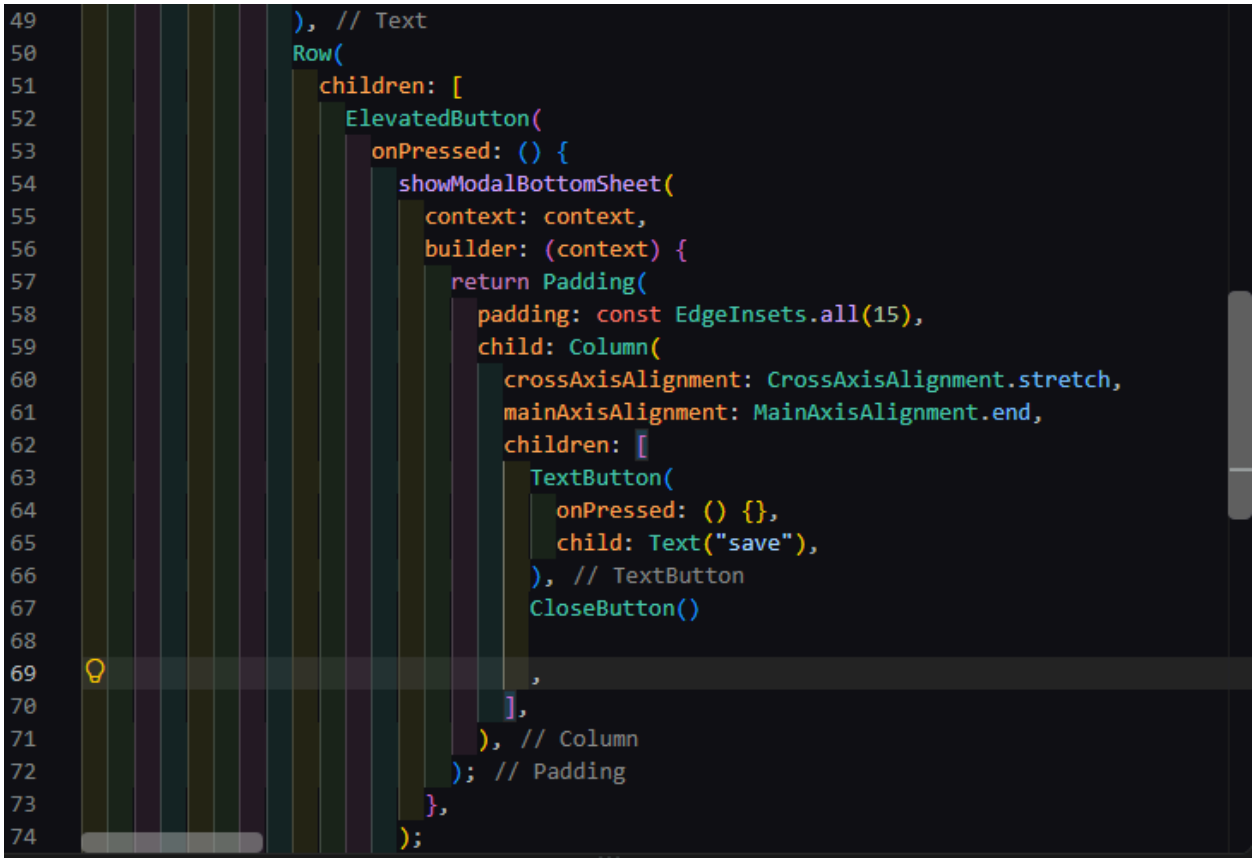
Test

1. Open Student Details.
2. Select Edit.
3. An editing modal should appear.
4. The modal should display the appropriate student's information.
5. Close or submit the modal.

Screenshot:

- Student Details Screen
- Edit modal
- Modal code


student information
DEBUG
student ID: IT-2023-009
name: siomai rice
year level: 3
course: BS Information Technology
hobby: Cycling
favorite subject: Object-Oriented Programming
Edit
save



```
), // Text
Row(
  children: [
    ElevatedButton(
      onPressed: () {
        showModalBottomSheet(
          context: context,
          builder: (context) {
            return Padding(
              padding: const EdgeInsets.all(15),
              child: Column(
                crossAxisAlignment: CrossAxisAlignment.stretch,
                mainAxisAlignment: MainAxisAlignment.end,
                children: [
                  TextButton(
                    onPressed: () {},
                    child: Text("save"),
                  ), // TextButton
                  CloseButton()

                  ,
                ],
              ), // Column
            ); // Padding
          },
        );
```

🏴 Flag 5 — Create the Add Student Screen

Objective: Create a separate screen for adding a new student.

Add an action from the Student List that navigates to:

Student List → Add Student

Create the **Add Student Screen**.

For now, focus on creating the screen and navigation structure.

Test

Selecting Add Student should open the Add Student Screen.

Screenshot:

- Student List with Add action
- Add Student Screen
- Navigation code

Add student

name: kulafu man

year level: 3

course: BS Information Technology

hobby: Playing basketball

favorite subject: Mobile Development



student ID: IT-2023-002

name: whiskey

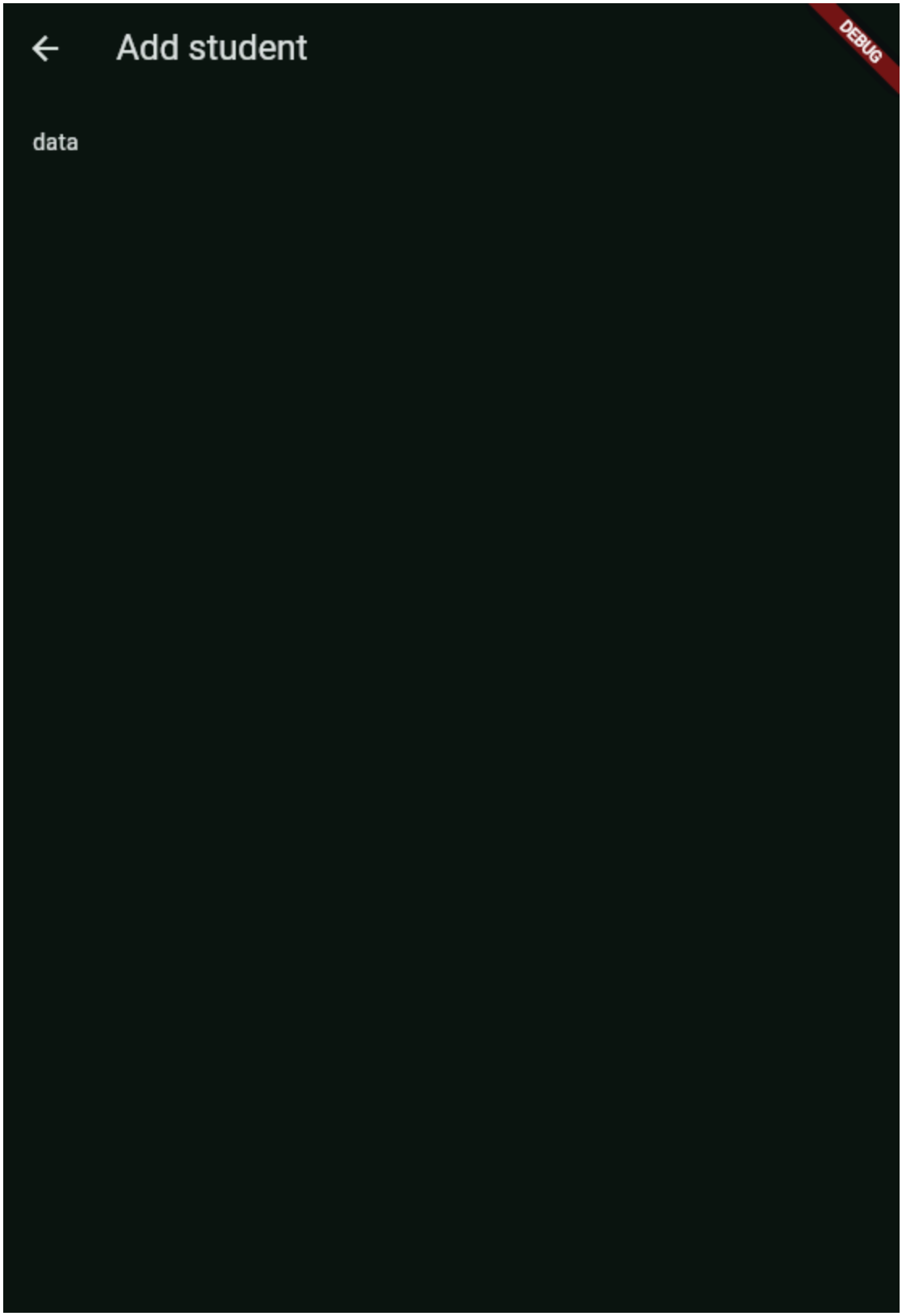
year level: 3

course: BS Information Technology

hobby: Reading novels

favorite subject: Web Systems and Technologies




Add student
DEBUG
data

```
body: Padding(
  padding: EdgeInsetsGeometry.all(16),
  child: Column(
    children: [
      ElevatedButton(
        onPressed: () {
          Navigator.push(
            context,
            MaterialPageRoute(
              builder: (context) {
                return AddStudentP();
              },
            ), // MaterialPageRoute
          );
        },
        child: Text("Add students"),
      ), // ElevatedButton
```

🏴 Flag 6 — Prepare the Add Student Screen

Objective: Prepare the Add Student Screen for future functionality.

Organize the screen so it is ready to receive student input.

Do not implement the student input fields yet.

The screen should simply exist as a separate destination.

Your application should now have:

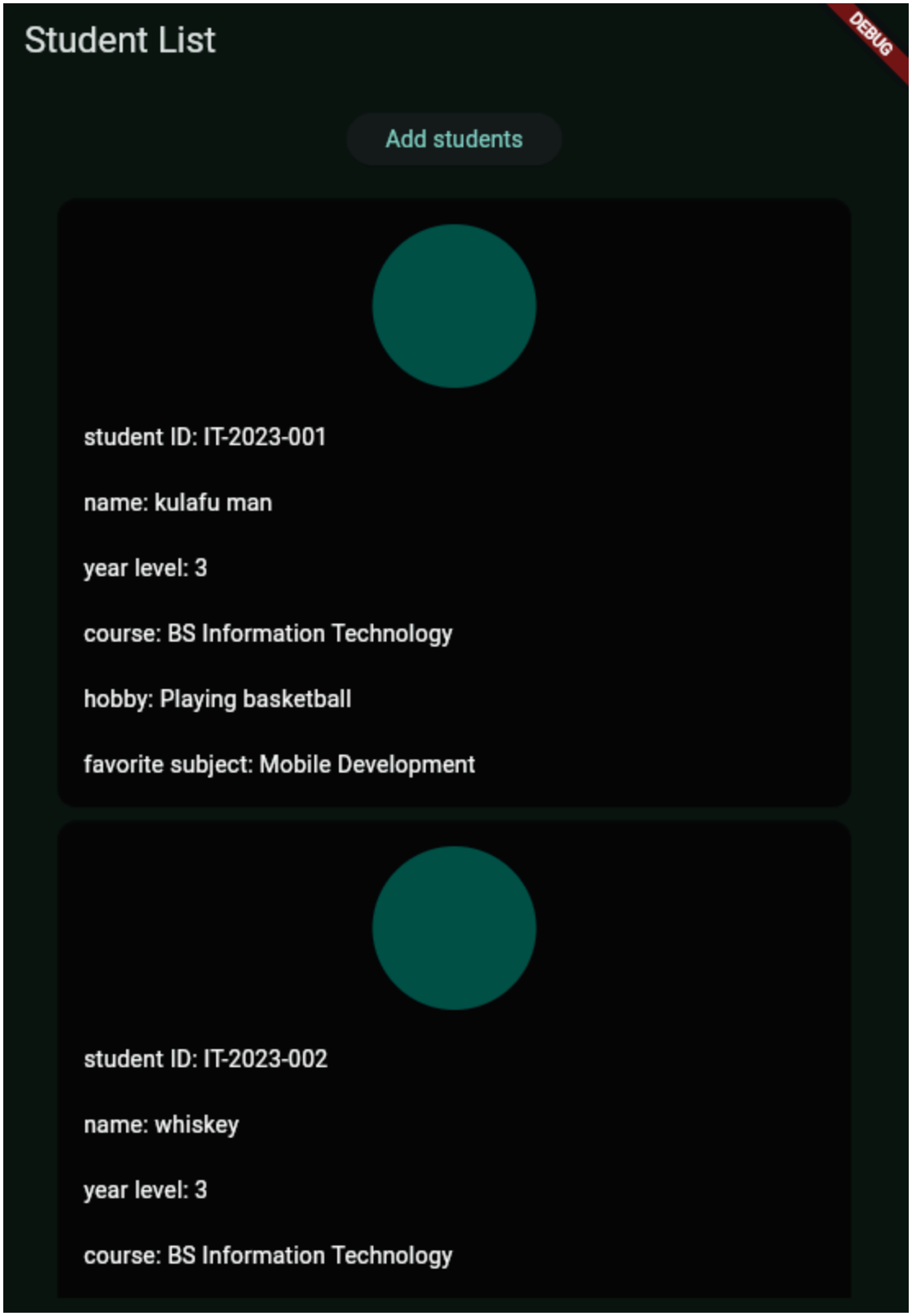
```
Student List
    ├── Student Details
    │       └── Edit Modal
    │
    └── Add Student
```

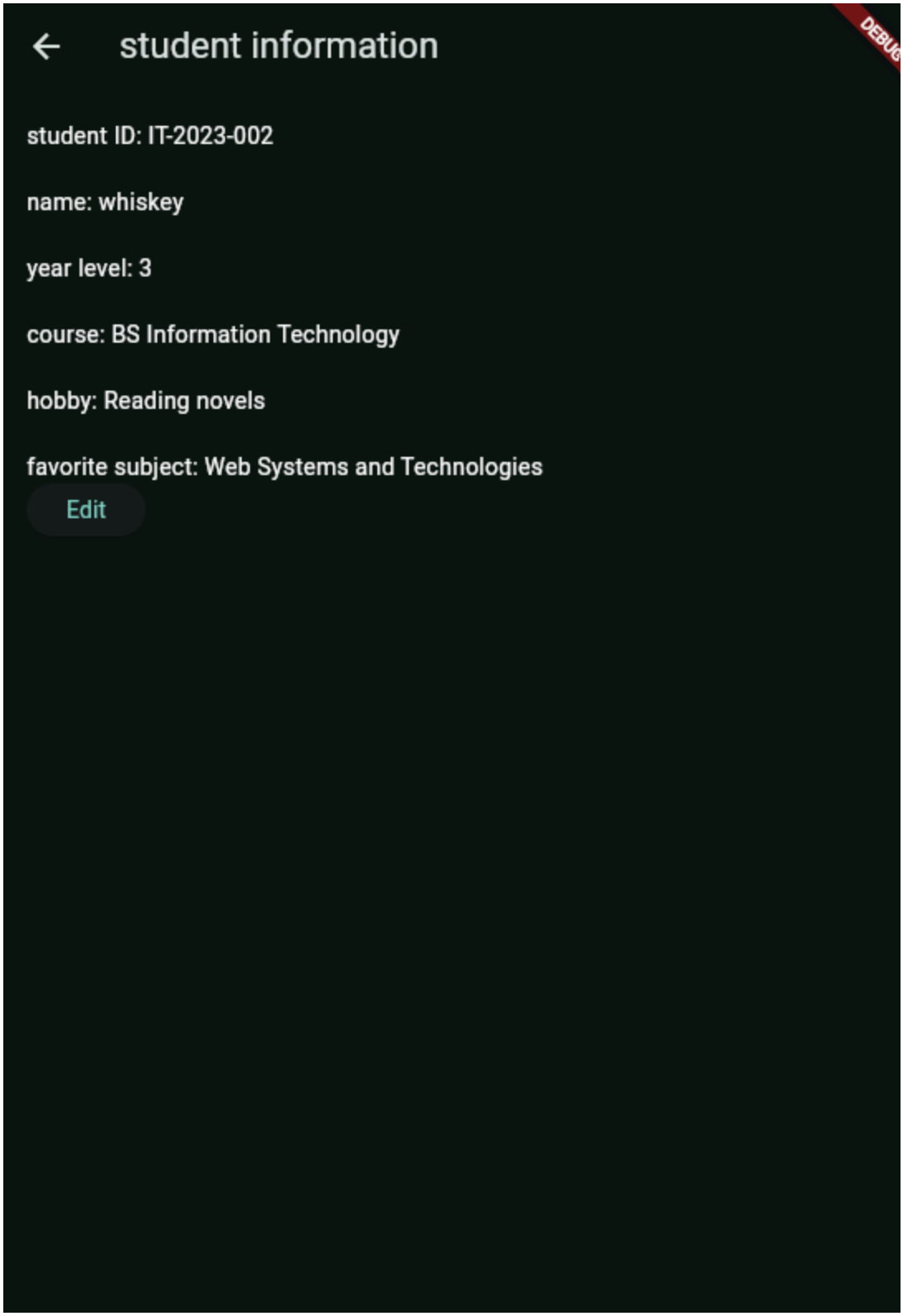
Test

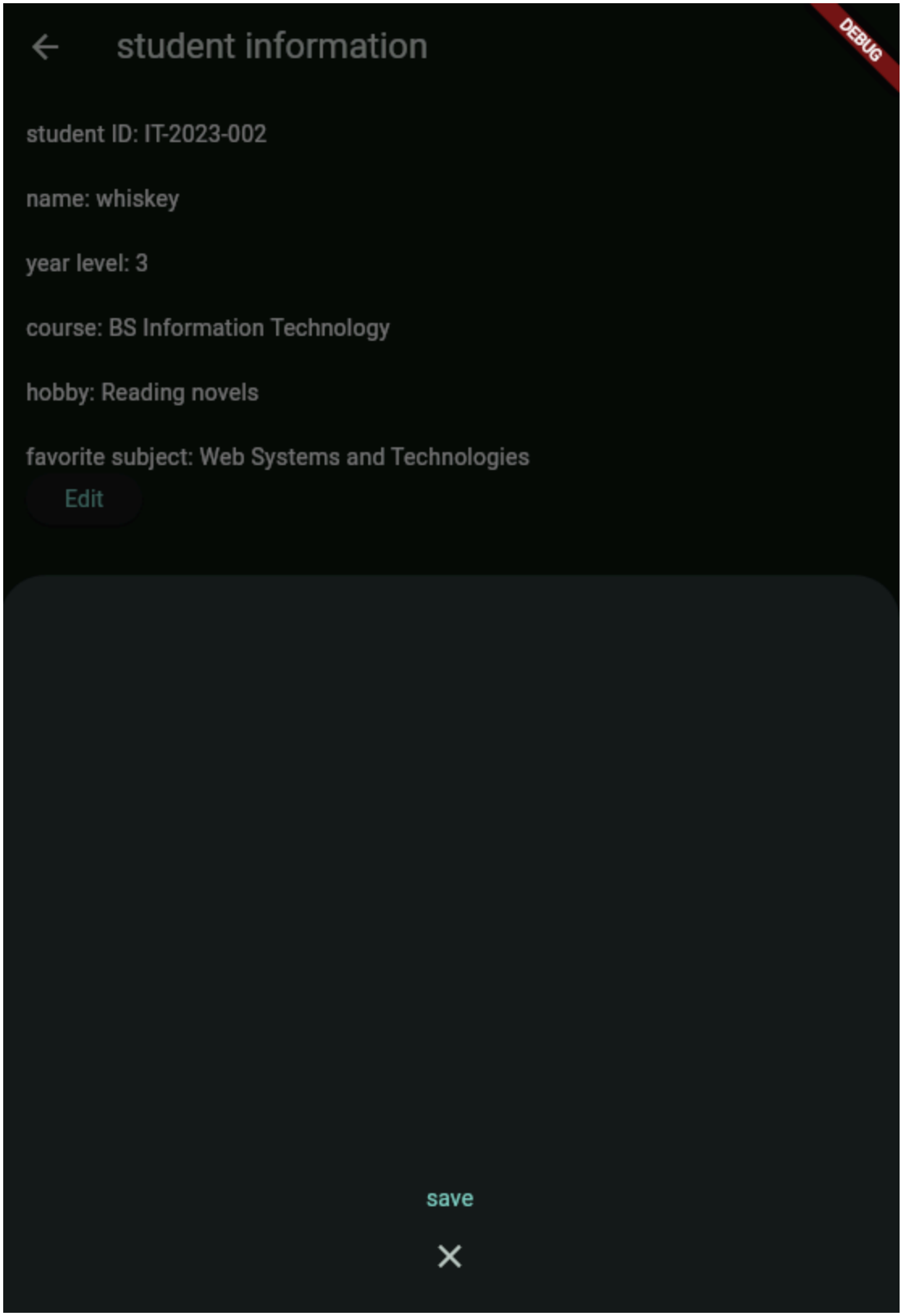
You should be able to navigate between the appropriate screens without placing everything on one screen.

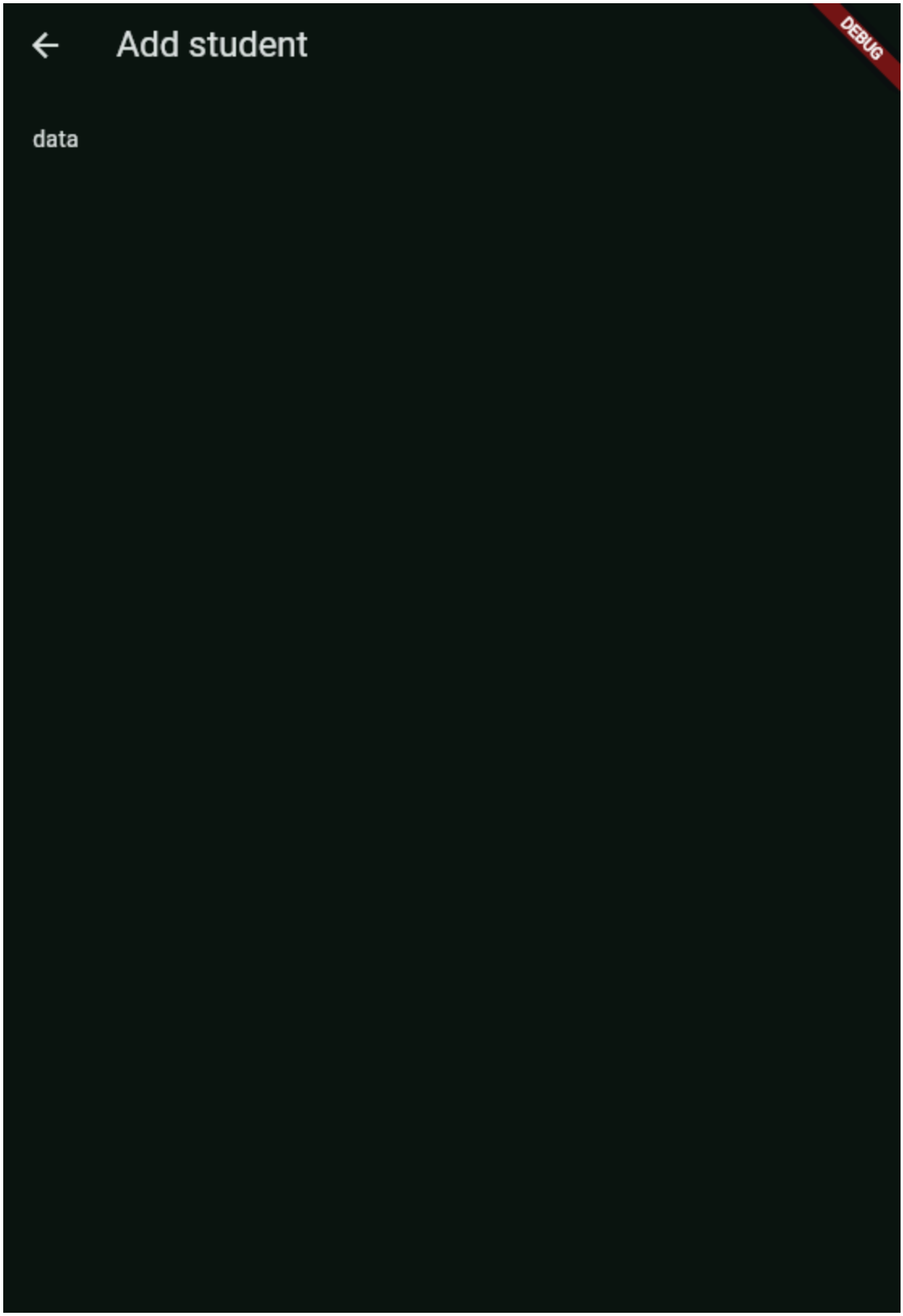
Screenshot:

- Student List
- Student Details
- Edit Modal
- Add Student Screen


Student List
DEBUG
Add students
student ID: IT-2023-001
name: kulafu man
year level: 3
course: BS Information Technology
hobby: Playing basketball
favorite subject: Mobile Development
student ID: IT-2023-002
name: whiskey
year level: 3
course: BS Information Technology


student information
DEBUG
student ID: IT-2023-002
name: whiskey
year level: 3
course: BS Information Technology
hobby: Reading novels
favorite subject: Web Systems and Technologies
Edit


student information
DEBUG
student ID: IT-2023-002
name: whiskey
year level: 3
course: BS Information Technology
hobby: Reading novels
favorite subject: Web Systems and Technologies
Edit
save



🏴 **Flag 7 — Connect the Routes**

Objective: Organize the application's screens using Flutter routes.

Create routes for the major screens:

/student-list
/student-details
/add-student

Research:

- Named routes
- `Navigator.pushNamed`
- Passing arguments through routes

The **Edit Modal should not be a route**.

Test

Verify that:

- Student List opens correctly.
- Student Details opens correctly.
- The correct student data is passed.
- Add Student opens correctly.
- Edit opens as a modal from Student Details.

Screenshot:

- Route configuration
- Navigation using routes
- Student Details receiving student data
- Edit modal

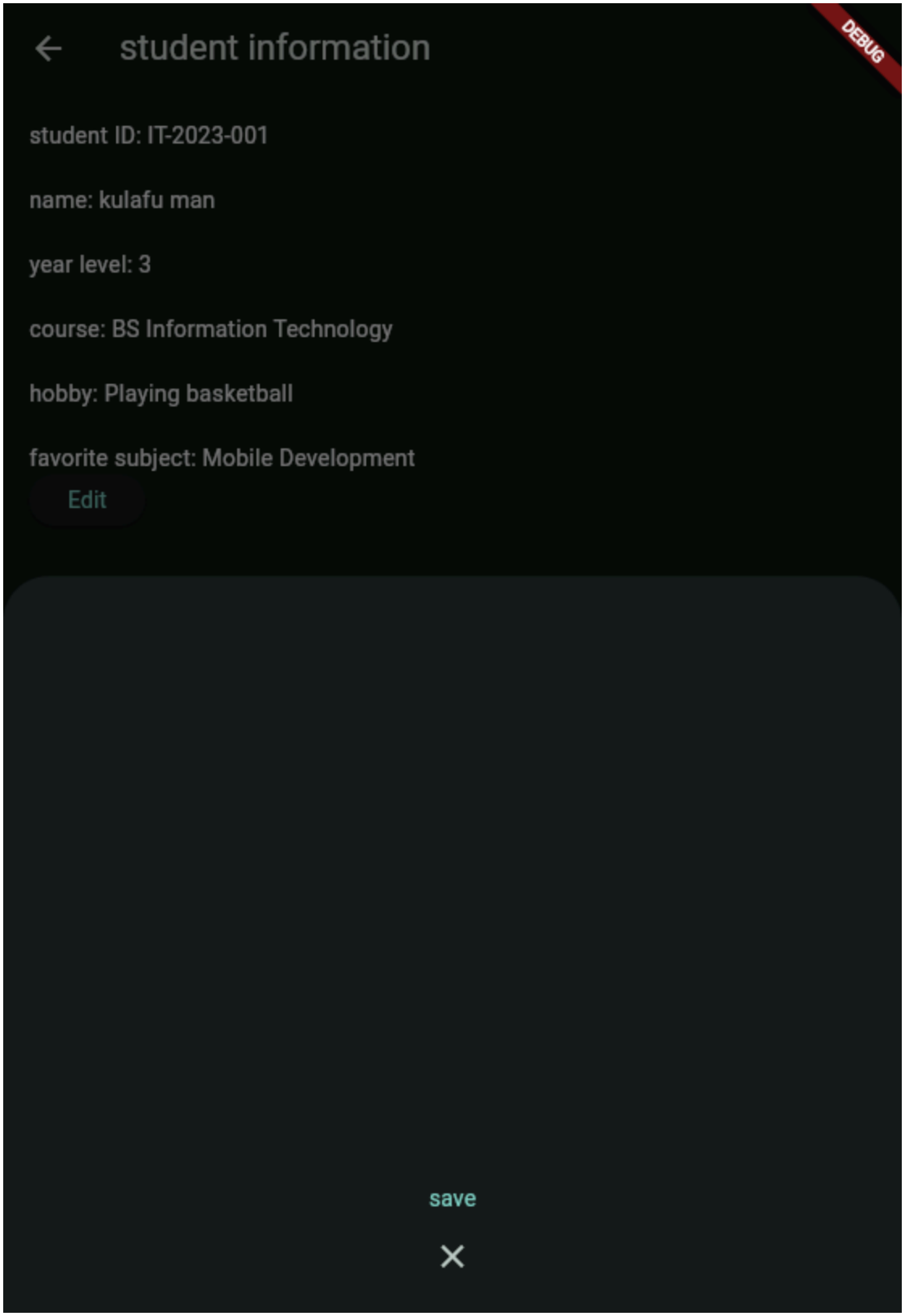
```dart
class MainApp extends StatelessWidget {
  final Student student;
  const MainApp({super.key, required this.student});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      initialRoute: "/",
      routes: {
        "/studentList": (context) => StudentListScreenW(student: student),
        "/student-details": (context) => StudentPage(student: student),
        "/add-student": (context) => AddStudentP(),
      },
      theme: ThemeData(
```

```dart
children: [
  ElevatedButton(
    onPressed: () {
      Navigator.pushNamed(context, "/add-student");
    },
    child: Text("Add students"),
  ), // ElevatedButton
```

```dart
class StudentPage extends StatelessWidget {
  final Student student;
  const StudentPage({super.key, required this.student});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("student information")),
      body: Padding(
        padding: EdgeInsetsGeometry.all(16),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.stretch
```


student information
DEBUG
student ID: IT-2023-001
name: kulafu man
year level: 3
course: BS Information Technology
hobby: Playing basketball
favorite subject: Mobile Development
Edit
save

🏴 Flag 8 — Save Your Progress

Objective: Commit and push your completed Multi-Screen Student Directory.

Use a proper Git commit message.

Branch:

`feature/multi-screen-navigation`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

